Tidewater Sync 2.4 – Command-line Guide

Tidewater Sync copies a directory tree to remote storage on a schedule. This guide explains how to install the tool, write its configuration file and diagnose the most common failures.

1. Installation

The tool needs Python 3.10 or later. Install it from the package index and check the version number afterwards.

```
$ pip install tidewater-sync
$ tidewater --version
tidewater 2.4.1
```

2. Configuration

2.1 The configuration file

By default the tool reads tidewater.yaml from the current directory. A minimal file looks like this:

```
# tidewater.yaml
source: /srv/data/exports
target: s3://example-bucket/backups
schedule: "0 2 * * *"
retention_days: 30
exclude:
  - "*.tmp"
  - ".cache/"
```

Every key is described in the table below. Unknown keys are rejected so that typing mistakes are caught early.

Table 1: Configuration options

Option	Type	Default	Description
source	path	—	Directory to read from. It must exist and be readable.
target	URL	—	Destination, for example a bucket or an SFTP path.
schedule	cron	"0 2 * * *"	When to run in daemon mode.
retention_days	integer	30	Days to keep old snapshots before they are deleted.
exclude	list	[]	Glob patterns that are skipped.

2.2 Environment variables

The following variables override the file and are useful in containers:

- TIDEWATER_CONFIG – path of the configuration file
- TIDEWATER_LOG_LEVEL – one of debug, info, warning or error
- TIDEWATER_DRY_RUN – set to 1 to list actions without copying

3. Running the tool

Run tidewater sync once to copy the changes since the last run, or tidewater daemon to keep running on the configured schedule. The process always finishes the file it is copying before it stops. A summary line is printed at the end of every run.

Table 2: Exit codes

Code	Meaning
0	Success
1	Configuration error
2	Network failure
3	Partial transfer

4. Troubleshooting

If a run fails, work through the following steps in order:

1. Run again with the log level set to debug.
   - Look for lines with "level": "error".
   - Note the retry counter of the failed upload.
2. Check that the target is reachable from the machine that runs the tool.
3. Verify the credentials and the clock of the machine, because expired signatures are rejected.

A failed upload is reported as a JSON record similar to the following one:

```
{
  "level": "error",
  "msg": "upload failed",
  "file": "reports/2026-03.csv",
  "retry": 3
}
```

If the problem persists, attach the log file to your support request together with the output of tidewater --version.